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Optimization of secure wireless communications for IoT networks in the presence of eavesdroppers

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ABSTRACT

The problem motivates this paper is that securing the critical data of 5G based wireless IoT network is of significant importance. Wireless 5G IoT systems consist of a large number of devices (low-cost legitimate users), which are of low complexity and under strict energy constraints. Physical layer security (PLS) schemes, along with energy harvesting, have emerged as a potential candidate that provides an effective solution to address this issue. During the data collection process of IoT, PHY security techniques can exploit the characteristics of the wireless channel to ensure secure communication. This paper focuses on optimizing the secrecy rate for simultaneous wireless information and power transfer (SWIPT) IoT system, considering that the malicious eavesdroppers can intercept the data. In particular, the main aim is to optimize the secrecy rate of the system under signal to interference noise ratio (SINR), energy harvesting (EH), and total transmits power constraints. We model our design as an optimization problem that advocates the use of additional noise to ensure secure communication and guarantees efficient wireless energy transfer. The primary problem is non-convex due to complex objective functions in terms of transmit beamforming matrix and power splitting ratios. We have considered both the perfect channel state information (CSI) and the imperfect CSI scenarios. To circumvent the non-convexity of the primary problem in perfect CSI case, we proposed a solution based on the concave-convex procedure (CCCP) iterative algorithm, which results in a maximum local solution for the secrecy rate. In the imperfect CSI scenario, we facilitate the use of S-procedure and present a solution based on the iterative successive convex approximation (SCA) approach. Simulation results present the validations of the proposed algorithms. The results provide an insightful view that the proposed iterative method based on the CCCP algorithm achieves higher secrecy rates and lower computational complexity in comparison to the other algorithms.

1. Introduction

Internet of Things (IoT) has emerged as an integrated part of the present communication system and has tremendous applications in security, logistics, quality control agriculture, etc. [1–3]. Wireless communication (e.g., wireless sensor network) is one of the key enabling technologies of IoT as it plays a significant role in data collection at the nodes. Security of the wireless medium is a great concern for communication systems because of their susceptibility to eavesdropping due to the nature of the wireless medium and its inherent characteristics [4]. Traditionally, the wireless communication systems rely on the application layer cryptographic algorithms for secure communication, however, due to the complexity of the encryption and decryption algorithms and limited computational capabilities of the edge points, it might not be a suitable choice for IoT [5,6]. In order to cover the computational complexity of the conventional higher layered cryptographic algorithms, physical layer security (PLS) algorithms have emerged as the potential solution. From the information theory perspective, PLS algorithms are capable of ensuring secure communication by exploiting the characteristics of the wireless channel.

Massive IoT in 5G is possible with the advancements in low power efficient communicating devices. Energy harvesting techniques, along with secrecy, will be a good choice for future IoT. Electromagnetic (EM) energy transfer techniques have emerged as one of the potential research areas over the energy crises. Simultaneous wireless information and power transfer (SWIPT) is a potential technique to overcome the bottleneck for providing sufficient power. In this method, the transmitter collectively transmits the energy signal along with the information signal [7–10]. SWIPT has been recognized as a promising technology to solve the energy scarcity problem in energy-constrained 5G wireless communication systems. The idea of SWIPT was first proposed by Varshney in [11], in which he characterizes the rate-energy trade-off in a discrete memory-less channel. Mainly there are two main techniques for the application of SWIPT, time switching (TS), and the power splitting (PS). In TS technique, the receiver switches between the information decoding (ID) and energy harvesting (EH) at any time [12,13], while in the later the receiver splits the signal into two streams, such that a fraction $\rho\,(0 \leq \rho \leq 1)$ of the received signal power is used for information decoding while the remaining $(1-\rho)$ is used for energy harvesting, respectively [14,15].

Focusing on this problem, secure communication in SWIPT systems has recently been pursued in a different context [16]. In SWIPT systems, we exploit the physical layer characteristics of the wireless

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channel such as fading and interference [17], to counter eavesdropping and ensure secure communication. Energy harvesting systems, specifically focus on beamforming with additional noise (AN) signal to ensure secure communication in wireless channels [18,19]. Wyner first introduced the concept of secrecy capacity in [20], in which he defined secrecy as the difference of mutual information between the legitimate channel and the wire-tap channel. Most of the previous work done in secure beamforming SWIPT system broadcast channels (BC) consider perfect channel state information (CSI) for the eavesdroppers (ED) as well as the users in the system [21–24]. However, due to the practical limitations, such as the quantization errors, delay errors, and limited feedback channel capacity, the transmitter is unable to get the perfect CSI. Recently, some potential work have been done, keeping into consideration the imperfect CSI for eavesdroppers, for secure SWIPT beamforming system [25–29]. To be more specific, [28] maximize the secrecy rate of the system model for SWIPT in the nonregenerative multi-antenna relay network with one energy receiver (ER), one information receiver (IR) and one eavesdropper. The authors proposed a constrained convex-concave procedure (CCCP) based on an iterative algorithm to maximize the secrecy rate and achieve a local optimum. Moreover, they proposed non-iterative semidefinite programming (SDP) based suboptimal solution and closed-form suboptimal solution. In [30], the authors proposed a suboptimal Gaussian randomization solution based on semidefinite relaxation (SDR) upper and lower bounds with one IR, one ER, and one ED, respectively. The authors in [29] take channel errors into account and proposed a worst-case secrecy rate maximization problem via SDP programming. They also focused on the stochastic uncertainty model and proposed a suboptimal solution for outage probability secrecy rate maximization based robust design. In [31], AN aided secrecy rate for multiple-input single-output (MISO) channel in the presence of multiple antennas eavesdroppers was investigated under both perfect and imperfect CSI. The author in [25] studied the max–min fairness EH among multiple multi-antennas energy receivers with channel errors considering power splitting. In [32], a SWIPT system with multiple antennas at ER and ID with perfect and imperfect CSI in the presence of AN was studied. The authors proposed a novel bisection search based reformulation of secrecy rate maximization problem into an associated power minimization problem. SWIPT in two-tier heterogeneous networks was studied in [27], for perfect CSI case two-stage problem based on SDR and one-dimensional line search were proposed, and successive convex approximation (SCA) approach has been applied for robust beamforming.

To summarize, research on SWIPT mainly focuses on traditional architectures, and the research for secrecy rate maximization for the SWIPT system with multiple users and eavesdroppers is still an open problem. SWIPT communication system with multiple users in the presence of eavesdroppers brings new challenges for secure joint beamforming and power splitting (JBPS) design. To the best of our knowledge, the security issue in JBPS design is still an open problem. This paper focuses on optimizing the secrecy rate of the wireless communication system, which consist of multiple transmit antennas, with K single antenna legitimate receivers and L single antenna eavesdroppers The desired user is assumed to adopt PS scheme to decode information and harvest energy, while the idle receivers can harvest energy at the same time. We study the design of the joint transmit beamforming vector for the desired user, with a combination of additional noise and power splitting ratio to maximize the secrecy rate. The main contributions of this paper are summarized below.

- Assuming perfect CSI, we first investigate the secure transmission issue for the SWIPT MISO system. Particularly, the transmit beamforming vectors and AN are jointly designed to optimize the secrecy rate of the system. The original problem is a non-convex due to the coupling of the optimization variable and is a hard optimization problem. In order to solve this non-convexity issue, we proposed a CCCP based iterative algorithm. By the application of this proposed algorithm, the original problem can be decomposed into subproblems. Each sub-problem in the iterative CCCP algorithm can be transformed into a second-order cone programming (SOCP) problem. The problem is guaranteed to converge to a locally optimal point.
- Further, we extend the secure beamforming optimization problem to the case of imperfect CSI, while considering the worst-case based optimization framework for secure communication. In this case, the beamforming vectors and AN is jointly designed to maximize the worst-case secrecy rate under transmit power constraint while guaranteeing the data transmit constraint, the energy harvesting constraints, and the worst-case transmission security constraint for eavesdroppers. We model imperfect CSI cases based on the deterministic norm bounded error CSI model. The problem is non-convex due to intractable semi-infinite constraints and the complex objective function. We facilitate the use of S- procedure for transforming semi-infinite constraints to linear matrix inequalities (LMI). The SCA approximation is applied to solve the non-convexity of the remaining constraint. The approximation can be refined at each iteration, and the local optimum of the original problem can be obtained.

The remainder of this paper is organized as follows. Section 2 presents the multi-user MISO system model and problem formulation. In Section 3, joint beamforming and AN designed is proposed with a perfect CSI scenario. Section 4 discusses the optimization design for the robust case. Finally, in Section 5 simulation results are presented to verify the proposed algorithm and conclusions are drawn in Section 6.

Notation. Boldfaced lower case and upper case letters represent vectors and matrices, respectively. $(\mathbf{A})^H$, $\|\mathbf{A}\|$ and $Tr(\mathbf{A})$ denote hermitian, Euclidean norm and trace of matrix $\mathbf{A}$, respectively. $\mathbb{R}^{m\times n}$ denotes the space of $m \times n$ real matrices and $\mathbb{R}_+$ denotes the set of positive real numbers. $\mathbf{A} \succeq 0$ denotes that $\mathbf{A}$ is a positive semi-definite matrix and $Rank(\mathbf{A})$ is the rank of matrix $\mathbf{A}$. $\mathbf{I}_{N\times N}$ denotes an $N\times N$ identity matrix. The operator vec(.) stacks the elements of a matrix in one long column vector, invp(x) denotes the inverse of the positive portion.

Notation	Definitation
N_T	The number of transmit antennas
K	The number of legitimate single antenna users
L	The number of single antenna eavesdroppers
$\mathbf{f}_k$	Beamforming vector for receiver k
$\mathbf{h}_k$	The channel between transmit antenna and k user
$\mathbf{g}_l$	The channel between transmit antenna and eavesdropper
s_k	The message intended for receiver k
n_s	The noise due to signal processing
$\mathbf{V}$	The noise covariance matrix
R_{sec}	The secrecy rate
E_k	The harvested energy of the k user
P_t	The total transmit power
ρ_k	The power splitting ratio
γ_k	The minimum target SINR at each receiver k
e_k	The required energy threshold at each receiver k
invp(a)	Inverse of a

2. System model

In this paper, the system is modeled as the SWIPT wireless communication system, as shown by Fig. 1. The system has N_T transmit

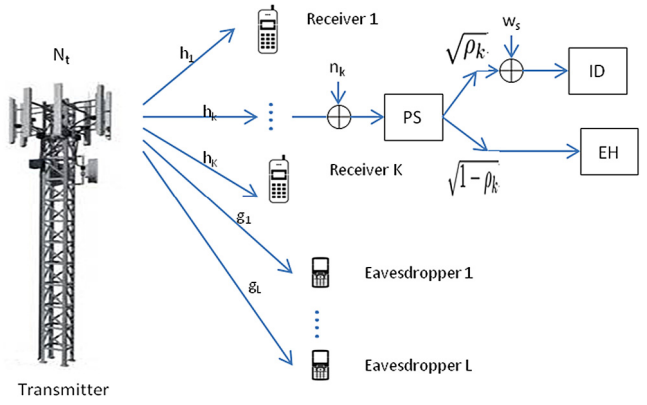


Fig. 1. System model.

antennas can be considered as the source in terms of the IoT network model, the source or the transmit antenna is capable of transmitting K independent message to K single antennas legitimate receivers which can relay simultaneously. In addition, L single antenna passive eavesdroppers are also present in the system. The wireless system from the relay to the destination is not considered in this particular scenario, but the same algorithm can be applied in terms of a single-hop wireless network. The direct link between the source and the destination is unavailable due to strong fading effects. The K independent legitimate receivers will act as the relay to forward messages from the sinking node to the destination node.

3. Problem formulation

The based band signal is expressed as

$$\mathbf{x} = \sum_{k=1}^{K} \mathbf{f}_k s_k + \mathbf{v} \tag{1}$$

where $s_k \in \mathbb{C}$ is the message intended for receiver k, $v \in \mathbb{C}^{N_T \times 1}$ denotes the additional noise vector and $\mathbf{f}_k \in \mathbb{C}^{N_T \times 1}$ is the corresponding beamforming vector for receiver k. The AN is modeled as $\mathbf{v} \sim \mathcal{CN}(0, \mathbf{V})$, where $\mathbf{V} = \mathbf{v}\mathbf{v}^H \succeq 0$ is the covariance matrix of $\mathbf{v}$. In the rest of the paper, we use $\mathcal{K} = 1, \ldots, K$ and $\mathcal{L} = 1, \ldots, L$, which denotes the index set of the receivers and the external eavesdroppers, respectively. Without loss of generality, we assume that $s_k \sim \mathcal{CN}(0, 1)$.

The received signal at the kth receiver and lth external eavesdropper are respectively given as

$$y_k = \mathbf{h}_k^H \mathbf{f}_k s_k + \mathbf{h}_k^H \sum_{j \neq k}^{K} \mathbf{f}_j s_j + \mathbf{h}_k^H \mathbf{v} + n_k \quad \forall k \in \{1, \ldots, K\} \tag{2}$$

$$y_l = \mathbf{g}_l^H \sum_{k=1}^{K} \mathbf{f}_k s_k + \mathbf{g}_l^H \mathbf{v} + n_l \quad \forall l \in \{1, \ldots, L\} \tag{3}$$

where $\mathbf{h}_k \in \mathbb{C}^{N_T \times 1}$ is the channel vector between the transmitter and the kth user, $\mathbf{g}_l \in \mathbb{C}^{N_T \times 1}$ denotes the channel vector between the transmitter and the l_{th} passive eavesdropper. It is noted here that the variable $\mathbf{h}_k$ and $\mathbf{g}_l$ includes the effects of the multi path, fading and path loss of the associated channels, respectively. $n_k \sim \mathcal{CN}(0, \sigma_k^2)$ and $n_l \sim \mathcal{CN}(0, \sigma_l^2)$ are modeled as additive white Gaussian noise (AWGN), for kth receiver and for lth eavesdropper, respectively. In addition, we assume that every receiver k exploits the power splitting scheme to handle the received signal i.e capable of denoting the information from the received signal and will harvest energy at the same time. Specifically, at receiver k, $\rho_k (0 \leq \rho_k \leq 1)$ portion of the signal power is consumed for information decoding and $1 - \rho_k$ is used for energy harvesting.

Therefore, we can write Eq. (2) as

$$y_k^{ID} = \sqrt{\rho_k}\left(\mathbf{h}_k^H \mathbf{f}_k s_k + \mathbf{h}_k^H \sum_{j \neq k}^{K} \mathbf{f}_j s_j + \mathbf{h}_k^H \mathbf{v} + n_k\right) + n_s \quad \forall k \in \{1, \ldots, K\} \tag{4}$$

where $n_s \sim \mathcal{CN}(0, \omega_s^2)$ is the additional noise caused by the signal processing.

The equivalent energy harvested signal at the receiver k can be expressed as

$$y_k^{ED} = \sqrt{1-\rho_k}\left(\mathbf{h}_k^H \mathbf{f}_k s_k + \mathbf{h}_k^H \sum_{j \neq k}^{K} \mathbf{f}_j s_j + \mathbf{h}_k^H \mathbf{v} + n_k\right) \quad \forall k \in \{1, \ldots, K\} \tag{5}$$

The signal to noise interference ratio (SINR) between the transmitter and idle receivers (potential eavesdroppers) k is given as [31]

$$SINR_k = \frac{\mathbf{f}_\mathbf{k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f}_\mathbf{k}}{\sum_{j \neq k}^{K} \mathbf{f}_\mathbf{j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f}_\mathbf{j} + Tr(\mathbf{h}_k \mathbf{h}_k^H \mathbf{V}) + \sigma_k^2 + \frac{\omega_s^2}{\rho_k}} \quad \forall k \tag{6}$$

On the other hand, the EH efficiency at the receiver k can be expressed as [32]

$$E_k = (1 - \rho_k)\eta\Big(\sum_{k=1}^{K} |\mathbf{h}_\mathbf{k}^\mathbf{H} \mathbf{w}|^2 + Tr(\mathbf{h}_k \mathbf{h}_k^H \mathbf{V}) + \sigma_k^2\Big), \quad \forall k \tag{7}$$

where $\eta \in (0, 1]$ denotes the EH efficiency at the receivers in converting the received radio signals to the electrical signals.

4. Secure beamforming with perfect CSI

The main is to optimize the secrecy rate of the wireless system. In this section, assuming perfect CSI, the focus is on jointly designing the beamforming vectors subject to the users SINR, EH and the transmit power constraints, in order to optimize the secrecy rate of energy harvested system. To begin with, the secrecy rate considering the worst case scenario is given by [33]

$$R_{sec} = \Big[log(1 + SINR_k) - \max_{l \in 1, \ldots, L} log(1 + SINR_l)\Big]^+ \tag{8}$$

The optimal resource allocation policy, $\{\mathbf{f}_k^*, \rho_k^*\}$, for maximizing the secrecy rate can be obtained by solving

$$\max_{\mathbf{f_k}, \mathbf{V}, \rho_k} \quad \min_{l \in \{1, \ldots, L\}} log \frac{\left(1 + \frac{\mathbf{f}_\mathbf{k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f}_\mathbf{k}}{\sum_{j \neq k}^{K} \mathbf{f}_\mathbf{j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f}_\mathbf{j} + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 + \frac{\omega_s^2}{\rho_k}}\right)}{\left(1 + \frac{\mathbf{f}_\mathbf{k}^H \mathbf{g}_l \mathbf{g}_l^H \mathbf{f}_\mathbf{k}}{\sum_{j \neq k}^{K} \mathbf{f}_\mathbf{j}^H \mathbf{g}_l \mathbf{g}_l^H \mathbf{f}_\mathbf{j} + \mathbf{g}_l^H \mathbf{V} \mathbf{g}_l + \sigma_l^2}\right)} \tag{9a}$$

$$\text{s.t} \quad SINR_k \geq \gamma_k, \ \forall k \tag{9b}$$

$$E_k \geq e_k, \ \forall k \tag{9c}$$

$$\sum_{k=1}^{K} |\mathbf{f}_k|^2 + Tr(\mathbf{V}) \leq P_t \tag{9d}$$

$$0 \leq \rho_k \leq 1 \ \forall k \in K \tag{9e}$$

where constraint (9b) is due to quality of service (QoS) which require that minimum SINR at each receiver k should be greater than the target value γ_k, constraint (9c) ensures that the minimum harvested energy threshold e_k should be satisfied at each receiver, (9d) ensures that the total minimum transit power is minimized while maximizing the secrecy rate of the system. Maximizing the secrecy rate in problem (9) is a non convex due to the non-convexity of the objective function and the coupling of optimization variable $\mathbf{f}_k$ and ρ_k in constraints (9b), (9c) and (9d). Since logarithmic function is monotonically increasing, therefore, it can be dropped from the objective function (9a). Problem (9) can be reformulated as

$$\max_{\mathbf{f_k}, \mathbf{V}, p_k, q_k, y, x, z} \quad z \tag{10a}$$

$$\text{s.t} \quad xy \geq z \tag{10b}$$

$$\left(1+\frac{\mathbf{f_k}^H\mathbf{g}_l\mathbf{g}_l^H\mathbf{f_k}}{\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{g}_l\mathbf{g}_l^H\mathbf{f_j}+\mathbf{g}_l^H\mathbf{V}\mathbf{g}_l+\sigma_l^2}\right)\leq(\frac{1}{y}) \tag{10c}$$

$$\left(1+\frac{\mathbf{f_k}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_k}}{\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2+\omega_s^2\rho_k}\right)\geq(x) \tag{10d}$$

$$\frac{1}{\gamma_k}\mathbf{f_k}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_k}\geq\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2+\omega_s^2p_k \tag{10e}$$

$$\sum_{k=1}^{K}|\mathbf{h_k^H}\mathbf{w}|^2+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2\geq\frac{e_kq_k}{\eta_k} \tag{10f}$$

$$\sum_{k=1}^{K}\mathbf{f}_k^H\mathbf{f}_k+Tr(\mathbf{V})\leq P_t \tag{10g}$$

$$p_k\geq1,q_k\geq1,invp(p_k)+invp(q_k)\leq1,\ \forall k,\quad x>0,\quad y>0 \tag{10h}$$

where $p_k=\frac{1}{\rho_k}$, $q_k=\frac{1}{1-\rho_k}$, x y and z in constraint (10a)–(10c) and (10h) are the slack variables, y represents the mutual information at the lth eavesdropper. In problem (10) it is easy to recognize the equivalence of problem (9) and (10) as the constraints in both of the problems will hold at equalities, when the optimal solution is obtained. A larger objective value can be obtained otherwise by increasing the values of the slack variables x, y and z. Therefore, we can deduce that the (10) is equivalent to the original problem (9). The formulated problem (10) is still non-convex. In order to circumvent the non-convexity issue, we proposed a solution that is motivated on the observation that problem (10) can be reformulated to a convex problem by the application of difference of convex (DC) programming. In DC programming, we make use of CCCP [34,35] algorithm to achieve local optimal solution, iteratively. The detailed procedure is explained below. In order to further simplify the algorithm, we define the following vectors:

$$\begin{aligned}&\mathbf{p}=[p_1,\ldots,p_k]^T,\quad \mathbf{q}=[q_1,\ldots,q_k]^T\\&\mathbf{f}=[\mathbf{f_1^T},\ldots,\mathbf{f_k^T}]^T,\quad \mathbf{r}=[\mathbf{p}^T,\mathbf{q}^T,\mathbf{f}^T,x,y]^T,\end{aligned} \tag{11}$$

Applying the vectors formed in Eq. (11) and rearranging the constraints problem (10) can be reformulated as

$$\max_{\mathbf{f_k},\mathbf{V},\mathbf{r}}\quad z \tag{12a}$$

$$\text{s.t} \quad xy \geq z \tag{12b}$$

$$\sum_{j=1}^{K}\mathbf{f_j}^H\mathbf{g}_l\mathbf{g}_l^H\mathbf{f_j}+\mathbf{g}_l^H\mathbf{V}\mathbf{g}_l+\sigma_l^2-v_l(\mathbf{r},\mathbf{V})\leq0 \tag{12c}$$

$$\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2+\omega_s^2p_k-x_k(\mathbf{r},\mathbf{V})\leq0 \tag{12d}$$

$$\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2+\frac{\omega_s^2(p_k+1)^2}{4}-y_k(\mathbf{r})\leq0 \tag{12e}$$

$$\frac{e_k(q_k+1)^2}{4}-\sigma_l^2-z_k(\mathbf{r},\mathbf{V})\leq0 \tag{12f}$$

$$\sum_{k=1}^{K}\mathbf{f}_k^H\mathbf{f}_k+Tr(\mathbf{V})\leq P_t \tag{12g}$$

$$x>0,\quad y>0,\quad p_k\geq1,q_k\geq1,invp(p_k)+invp(q_k)\leq1,\ \forall k \tag{12h}$$

where the last set of inequality constraint (12h) must be satisfied with equality at optimality, otherwise, the objective value can be further decreased by decreasing the p_k. $v_l(\mathbf{r},\mathbf{V})$, $x_k(\mathbf{r},\mathbf{V})$, $y_k(\mathbf{r})$ and $z_k(\mathbf{r},\mathbf{V})$ in constraint (12c)–(12f), respectively can be given as

$$v_l(\mathbf{r},\mathbf{V})\triangleq\frac{\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{g}_l\mathbf{g}_l^H\mathbf{f_j}+\mathbf{g}_l^H\mathbf{V}\mathbf{g}_l+\sigma_l^2}{y} \tag{13}$$

$$x_k(\mathbf{r},\mathbf{V})\triangleq\frac{\sum_{j=1}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\sigma_k^2+\omega_s^2p_k}{x} \tag{14}$$

$$y_k(\mathbf{r})\triangleq\frac{1}{\gamma_k}\mathbf{f_k}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_k}+\frac{\omega_s^2(p_k-1)^2}{4} \tag{15}$$

$$z_k(\mathbf{r},\mathbf{V})\triangleq\sum_{j=1}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k+\frac{e_k(q_k-1)^2}{4} \tag{16}$$

We remark that (13)–(16) are all convex functions jointly with respect to the variable in $\mathbf{r}\in\mathbb{R}_+^K\times\mathbb{R}_+^K\times\mathbb{C}_+^{KN_t\times1}\times\mathbb{R}_+\times\mathbb{R}_+$ and $\mathbf{V}$.

Proposition 1. *By using the concept of CCCP, problem (12) can be reformulated into the following DC problem (17).*

Proof. Please refer to Appendix A.

Thus, after the application of the CCCP algorithm, the problem (12) in its ith iteration can be formulated into the following convex optimization problem.

$$\max_{\mathbf{r},\mathbf{V},z}\quad z \tag{17a}$$

$$\text{s.t} \quad xy \geq z \tag{17b}$$

$$\sum_{j=1}^{K}\mathbf{f_j}^H\mathbf{g}_l\mathbf{g}_l^H\mathbf{f_j}+\sigma_l^2-\hat{v}_l(\mathbf{r}^{(i)},\mathbf{V}^{(i)},\mathbf{r},\mathbf{V})\leq0 \tag{17c}$$

$$\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\sigma_k^2+\omega_s^2p_k-\hat{x}_k(\mathbf{r}^{(i)},\mathbf{V}^{(i)},\mathbf{r},\mathbf{V})\leq0 \tag{17d}$$

$$\sum_{j\neq k}^{K}\mathbf{f_j}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{f_j}+\sigma_k^2+\frac{\omega_s^2(p_k+1)^2}{4}-\hat{y}_k(\mathbf{r}^{(i)},\mathbf{r})\leq0 \tag{17e}$$

$$\frac{e_k(q_k+1)^2}{4}-\sigma_l^2-\hat{z}_k(\mathbf{r}^{(i)},\mathbf{V}^{(i)},\mathbf{r},\mathbf{V})\leq0 \tag{17f}$$

$$\sum_{k=1}^{K}\mathbf{f}_k^H\mathbf{f}_k+Tr(\mathbf{V})\leq P_t \tag{17g}$$

$$x>0,\quad y>0,\quad p_k\geq1,q_k\geq1,invp(p_k)+invp(q_k)\leq1,\ \forall k \tag{17h}$$

whose solution is given by $\left(\mathbf{r}^{(i+1)},\mathbf{V}^{(i+1)}\right)$

Proposition 2. *By introducing the new sets of variables $\tilde{a}_l$, a_l, $\tilde{b}_k$, b_k, $\tilde{c}_k$, c_k, $\tilde{d}_k$ and d_k, $l\in\mathcal{L}$, $k\in\mathcal{K}$, problem (17) can be reformulated into the following SOCP problem (18).*

Proof. Proof is given in Appendix B.

It can also be proved that the proposed iterative based CCCP algorithm for optimizing the secrecy rate of $k\in\mathcal{K}$ users converges to a local optimal solution of (9). The proof is similar to that of Lemma 2 and Theorem 1 in [36] and is omitted for brevity. The proposed non-robust secrecy maximization algorithm is summarized in Algorithm 1.

$$\max_{\mathbf{r},\mathbf{V},z}\quad z \tag{18a}$$

$$\text{s.t} \quad \|[2z,x-y]\|\leq x+y \tag{18b}$$

$$\|[\mathbf{s}_l^T,\frac{\tilde{a}_l-a_l-1}{2}]\|\leq\frac{\tilde{a}_l-a_l+1}{2} \tag{18c}$$

$$\|[\mathbf{t}_k^T,\frac{\tilde{b}_k-b_k-1}{2}]\|\leq\frac{\tilde{b}_k-b_k+1}{2} \tag{18d}$$

$$\|[\mathbf{u}_k^T,\frac{\tilde{c}_k-c_k-1}{2}]\|\leq\frac{\tilde{c}_k-c_k+1}{2} \tag{18e}$$

$$\|[\sqrt{\frac{e_k}{4\eta_k}},\frac{\tilde{d}_k-d_k-1}{2}]\|\leq\frac{\tilde{d}_k-d_k+1}{2} \tag{18f}$$

$$\|[\mathbf{f}_1^T,\ldots,\mathbf{f}_K^T]\|+Tr(\mathbf{V})\leq P_t \tag{18g}$$

$$x>0,\quad y>0,\quad p_k\geq1,q_k\geq1,invp(p_k)+invp(q_k)\leq1,\ \forall k \tag{18h}$$

5. Robust secure beamforming with imperfect CSI

Its worth noting that the problem (10) considers a secure SWIPT system under the assumption of perfect CSI. In practice, during the transmission the legitimate receivers send feedback information to the

Algorithm 1 The Proposed CCCP-Based Iterative Algorithm

Inputs: Total transmit power P_t, beamforming vector $\mathbf{f}_k$, energy harvesting ratio p_k and q_k, energy harvesting efficiency at each receiver η_k, maximum number of iterations N_{max} and Eqs. (48), (49), (50) and (51).
Define the tolerance accuracy δ
Initialize the algorithm with a feasible point by solving problem (9) using SDP approach [25] to get $\mathbf{r}^{(0)}$, $\mathbf{V}^{(0)}$.
Set the iteration number $i = 0$
repeat
Compute th e affine approximation (13), (14), (15) and (16), respectively.
Solve problem (P5) and assign the solution to $\mathbf{r}^{(i+1)}$ and $\mathbf{V}^{(i+1)}$
Update the iteration number: $i = i + 1$
until Until convergence or max no if iteration is reached i.e $i > N_{max}$.
Outputs: Secrecy rate, slack variable z, vector $\mathbf{r}$ given by Eq. (11) and covariance matrix $\mathbf{V}$.

transmit antennas of successful transmission of data packets. As a result of which the transmit antennas are able to refine the channel estimate of the legitimate receivers. Therefore we can assume the perfect CSI for the legitimate receivers during the transmission. However, since there is no communication between the transmit antennas and the eavesdroppers that are present in the coverage area of the base station, the CSI of that particular eavesdroppers becomes outdated. To capture this effect, we consider a worst case imperfect CSI scenario for modeling the resulting CSI uncertainties. The erroneous channel model is based on the deterministic norm-bounded model [37].

Let $\hat{\mathbf{g}}_l$ denotes the estimated channel vector between the transmitter and the eavesdroppers l, respectively. The erroneous channel vector can be described as

$$\mathbf{g}_l = \hat{\mathbf{g}}_l + \Delta\mathbf{g}_l, \qquad \forall l \tag{19}$$

where $\Delta\mathbf{g}_l \in \mathcal{G}_l$ is the CSI error vectors unknown to the transmitter. We assume that the error vectors $\Delta\mathbf{g}_l$ is bounded in their Euclidean norm, that is $\|\Delta\mathbf{g}_l\| \leq \epsilon_l$. The error bounded sets corresponding to these error vectors are respectively defined as

$$\mathcal{G}_l = \{\mathbf{g}_l | \mathbf{g}_l = \hat{\mathbf{g}}_l + \Delta\mathbf{g}_l, \|\Delta\mathbf{g}_l\| \leq \epsilon_l\}, \qquad \forall k \tag{20}$$

The shape and size of $\mathcal{G}_l$ model the type of uncertainty on the estimated CSI, which is linked to the physical phenomena producing the CSI errors. It should be emphasized that the actual errors $\Delta\mathbf{g}_l$ is assumed to be unknown, while the corresponding upper bounds ϵ_l can be obtained using preliminary knowledge about the type of imperfections and/or coarse knowledge of the channel type and main characteristics. In this section, we will address the secure optimization problem by employing the channel error model described above.

The aim is to investigate the secrecy rate maximization problem considering the imperfect CSI for the eavesdroppers. To begin with, in the presence of channel errors for eavesdroppers, problem (10) can be equivalently written as

$$\max_{\mathbf{f}_k,\mathbf{V},z,p_k,q_k,y,x} \quad z \tag{21a}$$

$$\text{s.t} \quad xy \geq z \tag{21b}$$

$$\max_{\epsilon_l} \quad \left(1 + \frac{\mathbf{f_k}^H \mathbf{g}_l \mathbf{g}_l^H \mathbf{f_k}}{\sum_{j\neq k}^K \mathbf{f_j}^H \mathbf{g}_l \mathbf{g}_l^H \mathbf{f_j} + \mathbf{g}_l^H \mathbf{V} \mathbf{g}_l + \sigma_l^2}\right) \leq (\frac{1}{y}) \tag{21c}$$

$$\left(1 + \frac{\mathbf{f_k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_k}}{\sum_{j\neq k}^K \mathbf{f_j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_j} + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 + \frac{\omega_s^2}{\rho_k}}\right) \geq (x) \tag{21d}$$

$$\frac{1}{\gamma_k} \mathbf{f_k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_k} \geq \sum_{j\neq k}^K \mathbf{f_j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_j} + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 + \omega_s^2 p_k \tag{21e}$$

$$\sum_{k=1}^K |\mathbf{h_k^H w}|^2 + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 \geq \frac{e_k q_k}{\eta_k} \tag{21f}$$

$$\sum_{k=1}^K \mathbf{f}_k^H \mathbf{f}_k + Tr(\mathbf{V}) \leq P_t \tag{21g}$$

$$x > 0, \quad y > 0, \quad p_k \geq 1, q_k \geq 1, invp(p_k) + invp(q_k) \leq 1 \tag{21h}$$

The problem (21) is non-convex due to the infinite constraint in the CSI errors in $\mathbf{g}_l$ and the coupling of the optimization variable $\mathbf{f}_k$ and ρ_k. In order to solve the problem more efficiently, we introduce a slack variables u. The optimization problem can be reformulated as

$$\max_{\mathbf{f}_k,\mathbf{V},z,p_k,q_k,y,x,u,v} \quad z \tag{22a}$$

$$\text{s.t} \quad xy \geq z \tag{22b}$$

$$\min_{\epsilon_l} \sum_{j\neq k}^K \mathbf{g}_l^H \mathbf{f}_j \mathbf{f}_j^H \mathbf{g}_l + \mathbf{g}_l^H \mathbf{V} \mathbf{g}_l + \sigma_l^2 \geq yu \tag{22c}$$

$$\max_{\epsilon_l} \sum_{j=1}^K \mathbf{g}_l^H \mathbf{f}_j \mathbf{f}_j^H \mathbf{g}_l + \mathbf{g}_l^H \mathbf{V} \mathbf{g}_l + \sigma_l^2 \leq u \tag{22d}$$

$$\left(1 + \frac{\mathbf{f_k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_k}}{\sum_{j\neq k}^K \mathbf{f_j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_j} + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 + \frac{\omega_s^2}{\rho_k}}\right) \geq (x) \tag{22e}$$

$$\frac{1}{\gamma_k} \mathbf{f_k}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_k} \geq \sum_{j\neq k}^K \mathbf{f_j}^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{f_j} + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 + \omega_s^2 p_k \tag{22f}$$

$$\sum_{k=1}^K |\mathbf{h_k^H w}|^2 + \mathbf{h}_k^H \mathbf{V} \mathbf{h}_k + \sigma_k^2 \geq \frac{e_k q_k}{\eta_k} \tag{22g}$$

$$\sum_{k=1}^K \mathbf{f}_k^H \mathbf{f}_k + Tr(\mathbf{V}) \leq P_t \tag{22h}$$

$$x > 0, \quad y > 0 \quad p_k \geq 1, q_k \geq 1, invp(p_k) + invp(q_k) \leq 1 \tag{22i}$$

Now we will try to deal with the non-convex constraints in the problem (22). The function on the right hand side (RHS) of constraint (22c) is quasi convex. We will first deal with the quasi convex problem after that we will apply the S procedure [38] to deal with the infinite constraints for CSI errors. Inspired by the idea of successive convex approximation (SCA) [39,40] technique as already been used in [27], we will approximate the terms in the right hand side with their convex upper estimates. Assuming $\theta \geq 0$, we define the following function

$$c_\theta(a, b) = \frac{\theta}{2} a^2 + \frac{1}{2\theta} b^2$$

is always convex and upper estimate of the function $c(a, b) = ab$ for a fixed θ. According to SCA, the RHS yu of constraint 2 can be written as $k_{\theta_1}(y^{(i)}, u^{(i)})$, where $\theta_1 = \frac{\tilde{u}}{\tilde{y}}$. It is pertinent to note here that initial values i.e $u^{(i)}$, and $x^{(i)}$ are to be selected at random and can be updated by the optimal solution at each iteration. Now we will focus on the infinite constraints, by applying the channel uncertainty model in (19), constraint (22c) and (22d) can be given as

$$\min_{\epsilon_l}; \ (\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l)^H \left(\sum_{j\neq k}^K \mathbf{f}_j \mathbf{f}_j^H + \mathbf{V}\right)(\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l) + \sigma_l^2 \geq k_{\theta_1}(y, u) \tag{23}$$

$$\max_{\epsilon_l} (\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l)^H \left(\sum_{j=1}^K \mathbf{f}_j \mathbf{f}_j^H + \mathbf{V}\right)(\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l) + \sigma_l^2 \leq u \tag{24}$$

Since the left hand side of (23) and (24) contains ellipsoidal uncertainty regions, to make the problem more traceable, we introduce some more slack variables t_{1l} and t_{21}. The constraints in (23) and (24) can be transformed as

$$\min_{\epsilon_l} (\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l)^H \left(\sum_{j\neq k}^K \mathbf{F}_j + \mathbf{V}\right)(\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l) \geq t_{1l} \tag{25}$$

$$t_{1l} + \sigma_l^2 \geq k_{\theta_1}(y, u) \tag{26}$$

$$\max_{\epsilon_l}(\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l)^H \Big(\sum_{j=1}^{K} \mathbf{F}_j + \mathbf{V}\Big)(\hat{\mathbf{g}}_l + \Delta\mathbf{g}_l) \le t_{2l} \tag{27}$$

$$t_{2l} + \sigma_l^2 \le u \tag{28}$$

By applying the S-procedure, we can rewrite (25) and (27) into linear matrix inequality form as

$$\begin{bmatrix} \boldsymbol{\Delta}_1 + \lambda_l \mathbf{I}_{N_T} & \boldsymbol{\Delta}_1 \hat{\mathbf{g}}_l \\ \hat{\mathbf{g}}_l^H \boldsymbol{\Delta}_1 & \hat{\mathbf{g}}_l^H \boldsymbol{\Delta}_1 \hat{\mathbf{g}}_l - t_{1l} - \lambda_l \epsilon_l \end{bmatrix} \tag{29}$$

$$\begin{bmatrix} -\boldsymbol{\Delta}_2 + \mu_l \mathbf{I}_{N_T} & -\boldsymbol{\Delta}_2 \hat{\mathbf{g}}_l \\ -\hat{\mathbf{g}}_l^H \boldsymbol{\Delta}_2 & -\hat{\mathbf{g}}_l^H \boldsymbol{\Delta}_2 \hat{\mathbf{g}}_l + t_{2l} - \mu_l \epsilon_l \end{bmatrix} \tag{30}$$

where $\boldsymbol{\Delta}_1 = \sum_{j\neq k}^{K} \mathbf{f}_j \mathbf{f}_j^H + \mathbf{V}$ and $\boldsymbol{\Delta}_2 = \sum_{j=1}^{K} \mathbf{f}_j \mathbf{f}_j^H + \mathbf{V}$, while $\lambda_l \ge 0$ and $\mu_l \ge 0$ are the slack variables. The remaining constraint can be solved by using the CCCP and SOCP method as explained in Section 3. With the transformation problem (22) is equivalently represented as

$$\max_{\mathbf{f}_k, \mathbf{V}, z, p_k, q_k, y, u, t_{1l}, t_{2l}, \lambda_l, \mu_l} \quad z \tag{31a}$$

$$\text{s.t} \quad \|[2z, x - y]\| \le x + y \tag{31b}$$

$$t_{1l} + \sigma_l^2 \ge k_{\theta_1}(y, u) \tag{31c}$$

$$t_{2l} + \sigma_l^2 \le u \tag{31d}$$

$$(29) \text{ and } (30), \ \lambda_l \ge 0, \ \mu_l \ge 0 \tag{31e}$$

$$\|[\mathbf{t}_k^T, \frac{\tilde{b}_k - b_k - 1}{2}]\| \le \frac{\tilde{b}_k - b_k + 1}{2} \tag{31f}$$

$$\|[\mathbf{u}_k^T, \frac{\tilde{c}_k - c_k - 1}{2}]\| \le \frac{\tilde{c}_k - c_k + 1}{2} \tag{31g}$$

$$\|[\sqrt{\frac{e_k}{4\eta_k}}, \frac{\tilde{d}_k - d_k - 1}{2}]\| \le \frac{\tilde{d}_k - d_k + 1}{2} \tag{31h}$$

$$\|[\mathbf{f}_1^T, \ldots, \mathbf{f}_K^T]\| + Tr(\mathbf{V}) \le P_t \tag{31i}$$

$$x > 0, \quad y > 0 y \ge 0, \ \mathbf{V} \succeq 0, \tag{31j}$$

$$p_k \ge 1, \ q_k \ge 1, \ invp(p_k) + invp(q_k) \le k \tag{31k}$$

The constraint set of problem (31a) is convex and can be efficiently solved by using existing softwares, e.g., CVX [38]. Finally the robust algorithm is summarized in Algorithm 2. Let the optimal solution of problem (31a) is denoted by $(\mathbf{f}_k^*, \mathbf{V}^*, \rho_k^* \ \forall k)$.

We can show that the proposed CCCP based robust iterative algorithm for the beamforming design convergences to the local optimal solution of the problem (21). The proof is similar to the that of Lemma 3 and Theorem 1 in [36] and can also be found in [41], we therefore omit the details. We summarized the proposed robust case secrecy rate optimization algorithm in Algorithm 2.

Algorithm 2 Robust secure Iterative Algorithm

Inputs: Slack variable x, y and z, total transmit power P_t, noise covariance matrix $\mathbf{V}$, beamforming vector for each user $\mathbf{f}_k$, Eqs. (48), (49), (50) and (51), energy harvesting ratios p_k and q_k, total number of iterations N_{max}, error bound for eavesdropper channel ϵ_l.
Define the tolerance accuracy δ
Initialize the algorithm with a feasible point [25,27], $u^{(i)}$, $v^{(i)}$, $x^{(i)}$, $y^{(i)}$, $\mathbf{f}_k^{(i)}$ and $\mathbf{V}^{(i)}$.
Set the iteration number $i = 1$

repeat
Solve problem (P8) with initial values and assign the solution to $u^{(i+1)}$, $v^{(i+1)}$, $\mathbf{f}^{(i+1)}$, $\mathbf{V}^{(i+1)}$, $x^{(i+1)}$ and $y^{(i+1)}$
Update the iteration number: $i = i + 1$
until
The convergence or max no if iteration is reached i.e $i > N_{max}$.
Obtain the optimal solution $\mathbf{f}_k^*, \mathbf{V}^*, \rho_k^*, u^*, v^*, x^*$ and y^*
Outputs: Noise covariance matrix $\mathbf{V}$, beamforming vector for each user $\mathbf{f}_k$, secrecy rate, maximized value of harvested energy ratios.

Note: At each iteration, the values of the variables are updated. In particular the optimal value of the variables at nth iteration is always a feasible solution to $(n+1)$th iteration. Since the optimal values at the $(n+1)$th iteration become greater or equal to the values at nth iteration, therefore there will be a monotonically increase or decrease in the secrecy rate at each iteration. Due to the transmit power constraint, there is an upper bound on the secrecy rate, the convergence of the proposed algorithm is guaranteed, as verified by the simulation results.

6. Complexity analysis

In Sections 3 and 4, we have proposed the perfect and imperfect CSI case secrecy rate maximization problem. In this section we analyze the computational complexity analysis of the proposed algorithm. Moreover, we apply the same basic element of complexity analysis as in [42]. For ease of comparison, we assume that all transmitters are equipped with the same number of antennas i.e., $N_T = N$, $\forall k$. The complexity of the non-robust secrecy rate maximization is dominated by problem (18) AI_1 times, where I_1 is the iteration number. Since the complexity of solving $\hat{v}_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$, $\hat{x}_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$, $\hat{y}_k(\mathbf{r}^{(i)}, \mathbf{r})$ and $\hat{z}_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$ is negligible compared to solving problem (18). Problem (18) involves $KL + 3K + 2$ SOCs constraints, including $K + 1$ SOCs of dimension 3, KL SOCs of dimensions $KL + 2$, $2K$ SOCs of dimension $2K + 3$ and 1 SOC of dimension $KN + N^2 + 1$. The number of decision variable is of the order of $\mathcal{O}(KN + N^2 + 2K + 2)$.

The complexity of robust secrecy rate maximization is dominated by problem (31) AI_2 times, where I_2 is the iteration number. Algorithm 2 in problem (31) involves $KN + N^2 + 4K + 2$ variables. Problem P8 involves $2KL$ linear matrix inequality constraints of size $N + 1$ and $2K + 3$ SOCs constraints, including $K + 1$ SOCs constraints of dimension 3, $2K$ SOCs constraints of dimension $2K + 3$ and 1 SOC of size $KN + N^2 + 1$.

Proposed algorithm	Complexity order (suppressing the $\ln(1/\epsilon)$)
Non-robust SR	$AI_1(((KN + N^2 + 1)^2 + 2k(2K + 3)^2 + KL(KL + 2)^2 + (K + 1)3^2 + n_1^2)),$ $n_1 = \mathcal{O}(KN + N^2 + 2K + 2)$
Robust SR	$AI_2((2KL(N + 1)^3 + 2K(2K + 3)^2 + (K + 1)3^2 + (KN + N^2 + 1)^2 + n_2^2)),$ $n_2 = \mathcal{O}(KN + N^2 + 2K + 4L + 2)$

7. Simulation results

In this section, in order to evaluate the performance of the proposed MISO secure energy harvesting system, numerical results have been obtained by performing computer simulations. In simulations, we assume that the channel vectors are randomly generated from independent and identical Rayleigh fading distribution with unity variance. The nominal system configuration is defined by the following choice of parameters: $N_T = 6$, $K = 3$, $L = 2$, $\eta = 1$, $\sigma^2 = \sigma_k^2 = \sigma_l^2 = -30$ dBm, $\omega_s^2 = -20$ dBm unless specified otherwise. In addition, the SINR of each receiver k is $\gamma_k = 10$ dB and the energy harvesting threshold $e_k = -10$ dBm. In the implementation of the algorithm, the tolerance parameter is chosen as $\delta = 2\times10^{-3}$ and the maximum number of iteration in $N_{\max} = 20$. All the modeling and solution of the algorithms are performed using CVX [38] on a desktop Intel (i3-2100) CPU running at 3.1 GHz and 4 GB RAM. The performance of our proposed algorithm has been compared with that of the SDR based optimal solution of the problem, for both perfect CSI and the robust case. We have also compared the performance with No-AN scheme, for No-AN algorithm, the same designed method has been used to that of our proposed algorithm except we set $\mathbf{V} = 0$.

Fig. 2 presents the convergence of the proposed Algorithms 1 and 2. The convergence behavior of the proposed CCCP and the robust algorithm is demonstrated for different values of the transmit power.

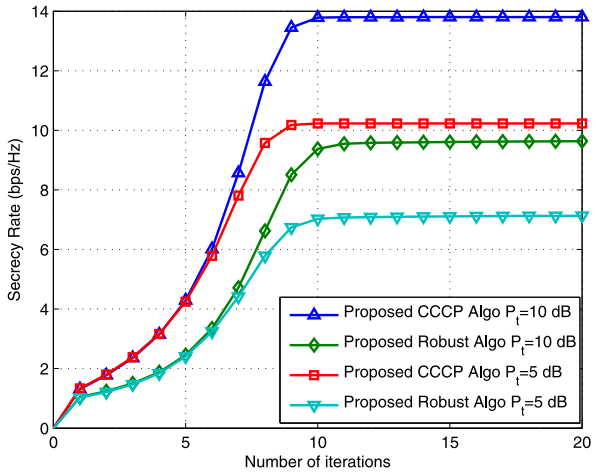


Fig. 2. Convergence property of the proposed algorithms according to different transmit power.

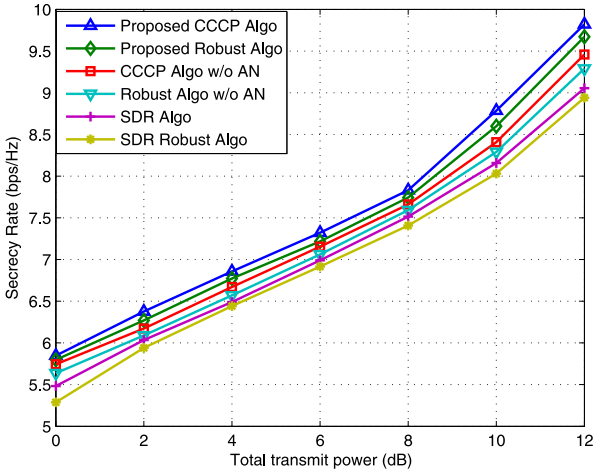


Fig. 3. Total transmit power versus secrecy rate with $N_T = 6$, $K = 3$, $L = 2$, $e_k = -10$ dBm and $\gamma_k = 10$ dB.

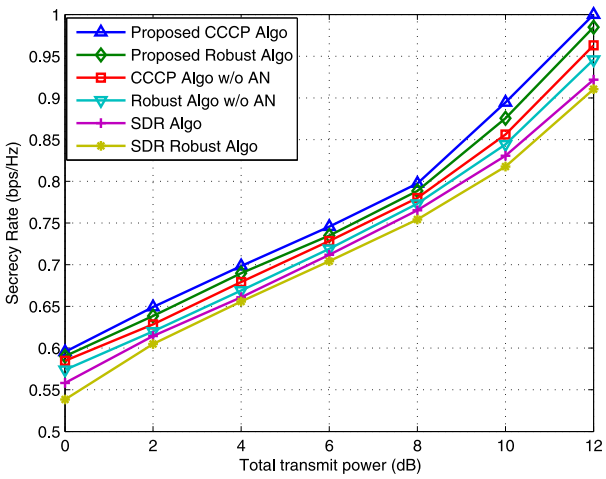


Fig. 4. Normalized transmit power versus secrecy rate with $N_T = 6$, $K = 3$, $L = 2$, $e_k = -10$ dBm and $\gamma_k = 10$ dB.

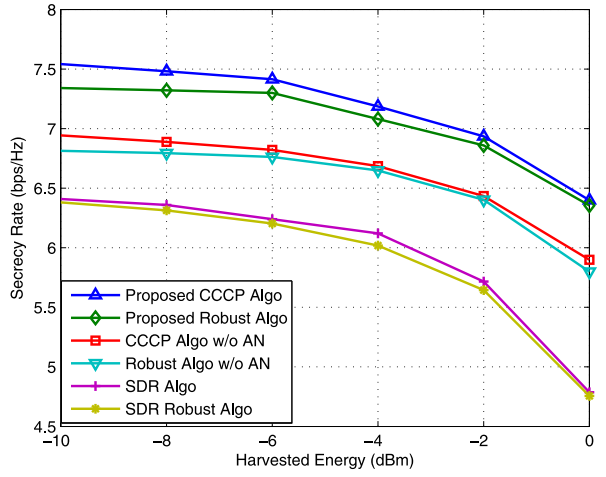


Fig. 5. Harvested energy e_k versus secrecy rate with $N_T = 6$, $K = 3$, $L = 2$, $P_t = 10$ dB and $\gamma_k = 10$ dB.

It is obvious that as we increase the transmit power constraint there is an increase in the secrecy rate of the system. It can also been shown that the secrecy rate of the proposed algorithm, as discussed before increase monotonically in each iteration and it after few iterations converges to a certain value.

Fig. 3 represents the performance comparison of the secrecy rate versus the total transmit power constraint with $N_t = 6$, $L = 2$ and $e_k = -10$ dBm. As it can been seen from Fig. 3, that our proposed CCCP and the robust algorithm outperforms SDR and No-AN schemes. The simulation results shows that proposed schemes achieves higher secrecy rates for all values of total transmit power. Moreover we also note higher secrecy rates can be obtained by applying our proposed algorithms as compared to the SDR algorithm.

Fig. 4 illustrates the normalized secrecy rate versus the transmitted total power constraint. It can be seen that as the total transmit power increases the performance gap between our proposed algorithm i.e CCCP for perfect CSI and robust algorithms increases as compared to the SDR based secrecy rate optimization method. Besides, it can be observed, that the CCCP and joint robust algorithm with even without additional noise performs better to the SDR approach for both robust and non-robust cases.

Fig. 5 plots the secrecy rate versus the harvested energy threshold constraint of each receiver with $N_t = 6$, $L = 2$ and $\gamma_k = -10$ dBm. It is clear from Fig. 5 that the secrecy rate for different algorithms decrease monotonically with increasing harvested energy e_k. This is attributed to the fact that there exist a trade-off between the energy harvested constraint at each receiver versus the secrecy rate. As we increase the harvested energy more power is consumed for EH of the user which results in lower secrecy rates. Moreover, a significant reduction is witnessed in the secrecy rate of the system with increasing energy harvested constraint which is attributed to the limited energy rate region. The CCCP and proposed robust algorithm achieves better performance gain in terms of secrecy rate as compared to the No-AN and SDR based schemes for all tested range of e_k. Fig. 6 depicts the total secrecy rate versus the number of transmit antennas with $L = 2$, $\gamma_k = 10$ dBm and $e_k = -10$ dBm. It is observed that the total secrecy rate of the system for all the schemes improves with increasing number of transmit antennas. The reason behind this fact is that as we increase the number of transmit antennas the extra degree of freedom is exploited for designing beamforming. Thus secrecy rate is directly proportional to the number of transmit antennas. Moreover, the secrecy rate for the No-AN design approaches to that our proposed algorithm when the number of antennas is large. The reason behind this interesting phenomena is that as degree of freedom increases the effect of additional noise is the system reduces.

Fig. 7 examines the effect of increasing eavesdroppers L on the secrecy rate of the system. It is assumed that $N_T = 9$, $K = 3$,

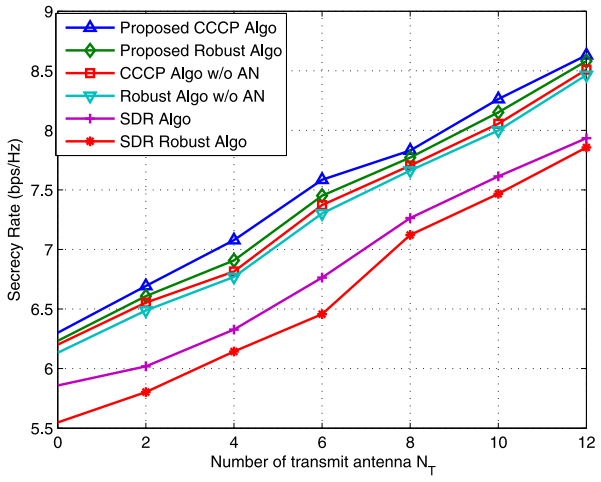


Fig. 6. Number of transmit antennas versus secrecy rate with $P_t = 10$ dB, $K = 3$, $L = 2$, $e_k = -10$ dBm and $\gamma_k = 10$ dB.

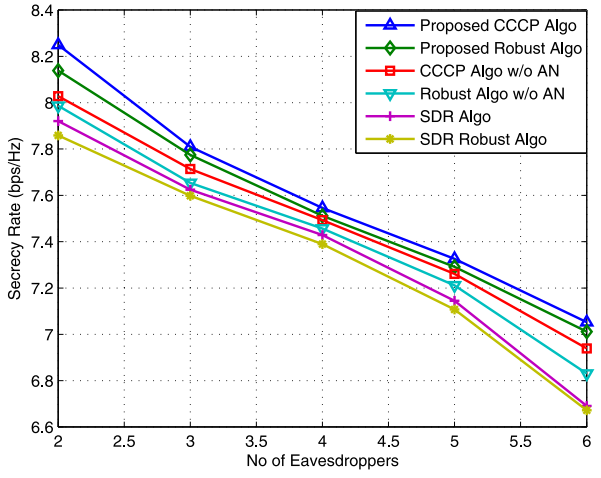


Fig. 7. No of eavesdroppers versus secrecy rate with $N_T = 6$, $K = 3$, $P_t = 10$ dB, $e_k = -10$ dBm and $\gamma_k = 10$ dB.

$\gamma_k = 10$ dBm and $e_k = -10$ dBm. It is observed from Fig. 7 that the total secrecy rate of the system for all the schemes decreases with the increase of eavesdropper L in the system. This is because as the number of eavesdropper increases more transmit power should be allocated to energy harvesting receivers and to generate AN in order to guarantee the EH and SINR constraint of the users. Allocating more power for generating the AN for jamming the channels of eavesdropper is difficult to achieve with the increasing number of eavesdroppers due to the transmit power constraint of the system, which result in reduce secrecy rate. Moreover it is clear from Fig. 7 that our proposed design outperform the benchmark and No-AN schemes for different number of EDs in the system.

8. Conclusion

This paper presents a novel algorithm to maximize the secrecy rate of the 5G wireless IoT system under both perfect and imperfect channel state information in the presence of malicious users. In order to maintain the QoS, the secrecy rate of the proposed algorithm is maximized, considering the SINR, energy harvesting, and the total transmits power constraints for each user. Additional noise is also added to prevent the system from malicious users. To address the nonconvexity issue of the secrecy rate maximization problem, we make use of the CCCP and SOCP algorithm fir the perfect CSI case. The proposed algorithm achieves a locally optimal solution to the problem. For a robust case, an iterative solution based on the SCA technique is proposed, which results in maximized secrecy. Simulation results are presented to validate the proposed algorithm.

9. Future work

In this section, we will be discussing future research directions. The channel error in this paper is only considered for eavesdroppers. In future work channels, errors will be considered for legitimate users as well. Feature of some emerging technologies in 5G wireless systems like directionality in mm Wave communication and superimposed signal in non-orthogonal multiple access techniques (NOMA), can be integrated to enhance the PLS of the 5G IoT network and need to be investigated. One of the main features of IoT is the easiness in the mobility of the user. Physical layer security issues need to be addressed in mobile users.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Muhammad Naeem Adil: Writing - review & editing. **MinJian Zhao:** Supervision.

Appendix A. The proof of Proposition 1

To proceed, we show that the problem (17) can equivalently written as a DC program, by using the concept of CCCP. Therefore problem (17) is an equivalent DC program. According to the concept of CCCP, we approximate the functions $\upsilon_l(\mathbf{r}, \mathbf{V})$, $x_k(\mathbf{r}, \mathbf{V})$, $y_k(\mathbf{r})$ and $z_k(\mathbf{r}, \mathbf{V})$ in the ith iteration by their first order Taylor expansion around the current point $\mathbf{r}^{(i)}$, denoted as $\hat{\upsilon}_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$, $\hat{x}_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$, $\hat{y}_k(\mathbf{r}^{(i)}, \mathbf{r})$ and $\hat{z}_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V})$, respectively. The approximated functions in (13)–(16) can be approximated by their Taylor series expansion given as

$$\hat{\upsilon}_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V}) = \upsilon_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) + 2R\{\nabla \upsilon_l(\mathbf{r}^{(i)})^H(\mathbf{r} - \mathbf{r}^{(i)})\} + 2R\{\nabla \upsilon_l(\mathbf{V}^{(i)})^H(\mathbf{V} - \mathbf{V}^{(i)})\} \tag{32}$$

$$\hat{x}_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V}) = x_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) + 2R\{\nabla x_k(\mathbf{r}^{(i)})^H(\mathbf{r} - \mathbf{r}^{(i)})\} + 2R\{\nabla x_k(\mathbf{V}^{(i)})^H(\mathbf{V} - \mathbf{V}^{(i)})\} \tag{33}$$

$$\hat{y}_k(\mathbf{r}^{(i)}, \mathbf{r}) = y_k(\mathbf{r}^{(i)}) + 2R\{\nabla y_k(\mathbf{r}^{(i)})^H(\mathbf{r} - \mathbf{r}^{(i)})\} \tag{34}$$

$$\hat{z}_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}, \mathbf{r}, \mathbf{V}) = z_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) + 2R\{\nabla z_k(\mathbf{r}^{(i)})^H(\mathbf{r} - \mathbf{r}^{(i)})\} + 2R\{\nabla z_k(\mathbf{V}^{(i)})^H(\mathbf{V} - \mathbf{V}^{(i)})\} \tag{35}$$

$$\nabla \upsilon_l(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) = \Big[\Big(0_{1\times(2K)}, \frac{(\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_\mathbf{1}^{(i)})^T}{y^{(i)}}, \ldots, \frac{(\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_{\mathbf{k-1}}^{(i)})^T}{y^{(i)}}, \frac{(\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_{\mathbf{k+1}}^{(i)})^T}{y^{(i)}}, \ldots, \frac{(\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_\mathbf{K}^{(i)})^T}{y^{(i)}}, 0_{1\times1}, -\frac{\sum_{j\neq k}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_j^{(i)} + \sigma_l^2}{2y^{(i)2}}\Big)^T, \mathbf{g}_l\mathbf{g}_l^H\Big] \tag{36}$$

$$\nabla x_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) = \Big[\Big(0_{1\times(k-1)}, \frac{\omega_s^2}{2x^{(i)}}, 0_{1\times(2K-k)}, \frac{(\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_\mathbf{1}^{(i)})^T}{x^{(i)}}, \ldots, \frac{(\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_K^{(i)})^T}{x^{(i)}}, -\frac{\sum_{j=1}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_\mathbf{j}^{(i)} + \sigma_k^2 + \omega_s^2 p_k^{(i)}}{2x^{(i)2}}, 0_{1\times1}\Big)^T, \mathbf{h}_k\mathbf{h}_k^H\Big] \tag{37}$$

$$\nabla y_k(\mathbf{r}^{(i)}) = [0_{1\times(k-1)}, \frac{\omega_s^2(p_k^{(i)} - 1)}{4}, 0_{1\times(2K-k)}, \frac{1}{\gamma_k}(\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_k^{(i)})^T, 0_{1\times2}]^T \tag{38}$$

$$\nabla z_k(\mathbf{r}^{(i)}, \mathbf{V}^{(i)}) = \Big[\Big(0_{1\times(k-1)}, \frac{e_k(q_k^{(i)}-1)}{4}, 0_{1\times(2K-k)}, (\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_1^{(i)})^T, \dots, (\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_1^{(i)})^T, 0_{1\times 2}\Big)^T, \mathbf{h}_k\mathbf{h}_k^H\Big] \tag{39}$$

where ∇ denotes the conjugate derivative of the respective function with respect to the complex vector $\mathbf{r}$ and $\mathbf{V}$. We note that $\upsilon_l(\mathbf{r}, \mathbf{V})$, $x_k(\mathbf{r}, \mathbf{V})$, $y_k(\mathbf{r})$ and $z_k(\mathbf{r}, \mathbf{V})$, are all affine function of $\mathbf{r}$ and $\mathbf{V}$.

Appendix B. The proof of Proposition 2

We first introduce the eight set of auxiliary variables $\tilde{a}_l$, a_l, $\tilde{b}_k$, b_k, $\tilde{c}_k$, c_k, $\tilde{d}_k$ and d_k, $l \in \mathcal{L}$, $k \in \mathcal{K}$, which satisfy

$$\tilde{a}_l = 2\mathcal{R}\left(\frac{\sum_{j\neq k}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_\mathbf{j}}{y^{(i)}}\right) - \left(\frac{\sum_{j\neq k}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{g}_l\mathbf{g}_l^H\mathbf{f}_j^{(i)} + \sigma_l^2}{y^{(i)2}}\right)y + \frac{\mathbf{g}_l^H\mathbf{V}\mathbf{g}_l}{y^{(i)}} \tag{40}$$

$$a_l = \sigma_l^2 - 2\frac{\sigma_l^2}{y^{(i)}} - \frac{\mathbf{g}_l^H\mathbf{V}^{(i)}\mathbf{g}_l}{y^{(i)}} \tag{41}$$

$$\tilde{b}_k = 2\mathcal{R}\left(\frac{\sum_{j=1}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_\mathbf{j}}{x^{(i)}}\right) + \frac{\mathbf{h}_k^H\mathbf{V}\mathbf{h}_k}{x^{(i)}} \left(\frac{\sum_{j=1}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_j^{(i)} + \sigma_k^2 + \omega_s^2 p_k^{(i)}}{x^{(i)2}}\right)x + \frac{\omega_s^2 p_k}{x^{(i)}} - \omega_s^2 p_k \tag{42}$$

$$b_k = \sigma_k^2 - \frac{\omega_s^2 p_k^{(i)}}{x^{(i)}} - \frac{2\sigma_k^2}{x^{(i)}} \tag{43}$$

$$\tilde{c}_k = \frac{2}{\gamma_k}\mathcal{R}\{\mathbf{f}_\mathbf{k}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_k\} + \frac{\omega_s^2 p_k(p_k^{(i)}-1)}{2} \tag{44}$$

$$c_k = \frac{1}{\gamma_k}\mathbf{f}_\mathbf{k}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_k^{(i)} + \sigma_k^2 - \frac{\omega_s^2(p_k^{(i)}-1)^2}{4} + \frac{\omega_s^2(p_k^{(i)}-1)p_k^{(i)}}{2} \tag{45}$$

$$\tilde{d}_k = 2\mathcal{R}\{\sum_{j=1}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_\mathbf{j}\} + \frac{e_k}{2\eta_k}(q_k^{(i)}-1)q_k + \mathbf{h}_k^H\mathbf{V}\mathbf{h}_k \tag{46}$$

$$d_k = \sum_{j=1}^{K}\mathbf{f}_\mathbf{j}^{(i)H}\mathbf{h}_k\mathbf{h}_k^H\mathbf{f}_j^{(i)} + \frac{e_k}{2\eta_k}(q_k^{(i)}-1)q_k^{(i)} - \frac{e_k(q_k^{(i)}-1)^2}{4\eta_k} + \sigma_k^2 \tag{47}$$

where a_l, b_k, c_k and d_k are constant scalars and $\tilde{a}_l$, $\tilde{b}_k$, $\tilde{c}_k$ and $\tilde{d}_k$ are affine functions in $\mathbf{r}$. The constraint in problem (18) can be rewritten in SOCP form as

$$\|[\mathbf{s}_l^T, \frac{\tilde{a}_l - a_l - 1}{2}]\| \leq \frac{\tilde{a}_l - a_l + 1}{2} \tag{48}$$

$$\|[\mathbf{t}_k^T, \frac{\tilde{b}_k - b_k - 1}{2}]\| \leq \frac{\tilde{b}_k - b_k + 1}{2} \tag{49}$$

$$\|[\mathbf{u}_k^T, \frac{\tilde{c}_k - c_k - 1}{2}]\| \leq \frac{\tilde{c}_k - c_k + 1}{2} \tag{50}$$

$$\|[\sqrt{\frac{e_k}{4\eta_k}}, \frac{\tilde{d}_k - d_k - 1}{2}]\| \leq \frac{\tilde{d}_k - d_k + 1}{2} \tag{51}$$

where

$$\mathbf{s}_l^T = [\mathbf{g}_l^H\mathbf{f}_1, \dots, \mathbf{g}_l^H\mathbf{f}_K]^T,$$
$$\mathbf{t}_k^T = [\mathbf{h}_k^H\mathbf{f}_1, \dots, \mathbf{h}_k^H\mathbf{f}_{k-1}, \mathbf{h}_k^H\mathbf{h}_{k+1}, \dots, \mathbf{h}_k^H\mathbf{f}_K]^T \textit{ and}$$
$$\mathbf{u}_k^T = [\mathbf{h}_k^H\mathbf{f}_1, \dots, \mathbf{h}_k^H\mathbf{f}_{k-1}, \mathbf{h}_k^H\mathbf{h}_{k+1}, \dots, \mathbf{h}_k^H\mathbf{f}_K, \frac{\omega_s(p_k+1)}{2}]^T$$

Therefore, (17) can be shown equivalent to the following SOCP problem

$$\begin{aligned} &\max_{\mathbf{r}} \quad z \\ &\text{s.t} \quad \|[2z, x-y]\| \leq x+y, \\ &\quad (48), (49), (50), (51), \\ &\quad \|[\mathbf{f}_1^T, \dots, \mathbf{f}_K^T]\| + Tr(\mathbf{V}) \leq P_t, \quad x>0, \quad y>0 \\ &\quad p_k \geq 1, q_k \geq 1, invp(p_k) + invp(q_k) \leq 1 \end{aligned}$$

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